

CSL reference

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1 Introduction

This is reference material for CSL. The Lisp identifiers mentioned here are the ones that are initially present in a raw CSL image. Some proportion of them are not really intended to be used by end-users but are merely the internal components of some feature.

2 Command-line options

The items shown here are the ones that are recognized on the CSL command line. In general an option that requires an argument can be written as either `-x yyy` or as `-xyyy`. Arguments should be case insensitive.

`--` * If the application is run in console mode then its standard output could * be redirected to a file using shell facilities. But the `--` * directive (followed by a file name) redirects output within the Lisp rather * than outside it. If this is done a very limited capability for sending * progress or status reports to stderr (or the title-bar when running in * windowed mode) remains via the `report!-right` function. * * The `-w` option may frequently make sense in such cases, * but if that is not used and the system tries to run in a window it will * create it starting off minimised.

`--cygwin` * On Windows this flag is used to specify that a cygwin rather than a native * windows version of Reduce is needed. This can be the situation in some * cases where the foreign function interface is to be used.

`--force-backtrace, --bt` * Forces any error to generate a backtrace regardless of any * attempt from with the system to change that (eg via use of `errorset`). * Intended for use during system debugging where it may be important to * observe behaviour otherwise hidden by (`errorset X nil nil`) but when it * undesirable to change the input script at all.

--force-echo * Forces echoing of input regardless of any * attempt from with the system to change that. Intended for use during system * debugging where it may be important to observe progress through an * input file but undesirable to change the input script at all.

--force-verbos * Forces generation of messages from the garbage collector regardless of any * attempt from with the system to change that. Intended for use during system * debugging where it may be important to see when a garbage collection occurs * but undesirable to change the input script at all.

--gc-error * `-gc-error=N` raises an error the start of GC number N.

--gc-every * `-gc-every=N` forces a GC every N conses (if N!=0).

--gc-stop * `-gc-stop=N` stops at the start of GC number N.

--gc-trace * `-gc-trace` leads to copious debugging trace output from garbage collection. * `-gc-trace=N` only starts that for GC number N and beyond.

--gc-trigger * `-gc-trigger=NNNN` causes a garbage collection to be forced on the NNNNth * occasion when that could possibly happen. This may sometimes be relevant * when trying to track down garbage collection related bugs. If the option * is set every garbage collection displays the count that it corresponds to, * and the intent is that these counts should be fairly deterministic.

--gui * Encourage the system to run in its own window. Similar behaviour * to `-w+`.

--guimin * Encourage the system to run as in its own window, but * start that window off minimised. Similar * to `-w..`.

--help * It is probably obvious what this option does! Note that on Windows the * application was linked as a windows binary so it carefully creates a * console to display the help text in, and organizes a delay to give * people a chance to read it.

--ignore-load-time * By default the time spend loading or autoloading optional * modules is accounted by CSL as part of "garbage collector time". * If this option is set it is just not recorded at all.

--kara * This it is intended for use by those maintaining CSL not for the general * public. By default long multiplication can use a threaded implementation * (to exploit multi-core machines). This happens when numbers get bigger * than some magic threshold. * This option allows one to override the default threshold so that * performance effects can be measured and the cut-off adjusted to suit the * machine involved.

`--kmax` * Normally this code goes back to the operating system and requests more memory * at any time when it feels that would be useful. This option can be used to * set an approximate limit on the amount it will use. See also `-k`

`--lisp` * This option causes Reduce to start up in symbolic mode.

`--maxmem` * Normally this code goes back to the operating system and requests more memory * at any time when it feels that would be useful. This option can be used to * set an approximate limit on the amount it will use. See also `-k`

`--mem` * See `"-k"` which behaves exactly the same.

`--no-rcfile` * Instruct the system not to read a user-specific configuration file, which * make have such a names as `".reducerc"`, at startup time. This * can be useful during system building where utterly self-contained and * predictable behaviour is important.

`--nogui` * Encourage the system to run as a console-style application. Similar * to `-w-` or just simply `-w`.

`--texmacs` * If CSL/Reduce is launched from texmacs this command-line flag should be * used to arrange that the `texmacs` flag is set in * `lispsystem!*`, and the code may then do special things.

`--trace`, `--tr` * When followed by the name of a function this command-line option has and * effect as if `(trace '(fname))` had been called at system start-up so that * all calls to the named function are reported to the user. Perhaps often to * be combined with `-bt` so that on any error a backtrace will get generated, * and used when an input script leads to failure and one wants to investigate * its behaviour without altering the script at all.

`--version` * It is probably obvious what this option does! But note that there is an * issue here about whether one is looking at the CSL version identification or * one for the Lisp application from an image file (eg often Reduce).

item `[ttfamily -wait]` `indexttfamily -wait` * This displays the process number and waits for 15 seconds at the * start of a run. This may be useful for those who have built everything * with debugging options and then want to start it fairly normally and * then attach from gdb or some other debugger.

`--waste` * `-waste N` performs N useless CONS operations during startup * so that memory layout and GC patterns are changed from the * default

- a * -a is a curious option, not intended for general or casual use. * If given it causes the (**batchp**) function to return the opposite * result from normal! Without “attfamily -a” (**batchp**) returns * T either if at least one file was specified on the command line, * or if the standard input is “not a tty” (under some operating systems this * makes sense – for instance the standard input might not be a “tty” if it * is provided via file redirection). Otherwise (ie primary input is directly * from a keyboard) (**batchp**) returns **nil**. Sometimes * this judgement about how “batch” the current run is will be wrong or * unhelpful, so -a allows the user to coax the system into better * behaviour. I hope that this is never used!

- b * -b tells the system to avoid any attempt to recolour prompts * and input text. It will mainly be needed on X terminals that have been * set up so that they use colours that make the defaults here unhelpful. * Specifically white-on-black and so on. * -b can be followed by colour specifications to make things yet * more specific. It is supposed to be the idea that three colours can be * specified after it for output, input and prompts, with the letters KRGYbMCW * standing for black, Red, Green, Yellow, blue, Magenta, Cyan and White. * This may not fully work yet!

- c * Displays a notice relating to the authorship of CSL. Note that this * is an authorship statement not a Copyright notice, because if any * (L)GPL code is involved that would place requirements on what was * displayed in a Copyright Notice.

- d * A command line entry -Dname=value or -D name=value * sets the value of the named lisp variable to the value as a string.

- dd * A command line entry -Dname=value or -DD name=value * sets the value of the named lisp variable to the value, interpreted as a * name, number, string etc using normal Lisp conventions. Cf -D which always * leaves the value as a string.

- e * If you follow ”-e” with a Lisp form then that gets evaluated and * its result printed during startup. This may be useful for forcing * settings or other debugging.

- f * At one stage CSL could run as a socket server, and -f portnumber * activated that mode. -f- used a default port, 1206 (a number * inspired by an account number on Titan that I used in the 1960s). The code * that supports this may be a useful foundation to others who want to make a * network service out of this code-base, but is currently disabled.

- g * In line with the implication of this option for C compilers, this enables * a debugging mode. It sets a lisp variable `!*backtrace` and * arranges that all backtraces are displayed notwithstanding use of * `errorset`.
- h * This option is a left-over. When the X-windows version of the code first * started to use Xft it viewed that as optional and could allow a build even when * it was not available. And then even if Xft was detected and liable to be used * by default it provided this option to disable its use. The remnants of the * switch that disabled use of Xft (relating to fonts living on the Host or * the Server) used this switch, but it now has no effect.
- i * CSL and Reduce use image files to keep both initial heap images and * “fasl” loadable modules. By default if the executable launched has some name, * say xxx, then an image file xxx.img is used. But to support greater * generality -i introduces a new image, -i- indicates * the default one and a sequence of such directives list image files that are * searched in the order given. These are read-only. The similar option * -o equally introduces image files that are scanned for input, but * that can also be used for output. Normally there would only be one * -o directive.
- j * Follow this directive with a file-name, and a record of all the files read * during the Lisp run will be dumped there with a view that it can be included * in a Makefile to document dependencies.
- k * -K `nnn` sets the size of heap to be used. If it is given then that much * memory will be allocated and the heap will never expand. Without this * option a default amount is used, and (on many machines) it will grow * if space seems tight. * * The extended version of this option is -K `nnn/ss` and then `ss` is the * number of “CSL pages” to be allocated to the Lisp stack. The default * value (which is 1) should suffice for almost all users, and it should * be noted that the C stack is separate from and independent of this one and * it too could overflow. * * A suffix K, M or G on the number indicates units of kilobytes, megabytes or * gigabytes, with megabytes being the default. So -K200M might * represent typical usage for common-sized computations. In general CSL * will automatically expand its heap, and so it should normally never be * necessary to use this option. It is legal to specify floating point * values such as -K1.2G.
- l * This is to send a copy of the standard output to a named log file. It is * very much as if the Lisp function (`spool ‘logfile’`) had been * invoked at the start of the run.
- m * Not used at present.

- n * Normally when the system is started it will run a “restart function” as * indicated in its heap image. There can be cases where a heap image has been * created in a bad way such that the saved restart function always fails * abruptly, and hence working out what was wrong becomes hard. In such cases * it may be useful to give the -n option that forces CSL to * ignore any startup function and merely always begin in a minimal Lisp-style * read-eval-print loop. This is intended for experts to do disaster recovery * and diagnosis of damaged image files.
- o * See -i. This specifies an image file used for output via * **faslout** and **reserve**.
- p * If a suitable profile option gets implemented one day this will activate it, * but for now it has no effect.
- q * This option sets **!*echo** to **nil** and switches off * garbage collector messages to give a slightly quieter run.
- r * The random-number generator in CSL is normally initialised to a value * that is intended not to be reproducible from run to run. * In many cases that behaviour is desirable, but for debugging it can be useful * to force a seed. The directive -r **nnn** sets the seed to * up to 64 bits taken from the value **nnn**. * and specifying -r0 explicitly asks for the non-reproducible * behaviour (I hope). Note that the main Reduce-level random number source is * coded at a higher level and does not get reset this way – this is the * lower level CSL generator.
- s * Sets the Lisp variable **!*plap** and hence the compiler generates * an assembly listing.
- t * -t **name** reports the time-stamp on the named module, and then * exits. This is for use in perl scripts and the like, and is * needed because the stamps on modules within an image or * library file are not otherwise instantly available. * * Note that especially on windowed systems it may be necessary to use this * with -- **filename** since the information generated here goes to * the default output, which in some cases is just the screen.
- u * See -d, but this forcibly undefines a symbol. There are probably * very very few cases where it is useful since I do not have a large * number of system-specific predefined names.
- v * An option to make things mildly more verbose. It displays more of a banner * at startup and switches garbage collection messages on.
- w * On a typical system if the system is launched it creates a new window and uses * its own windowed intarface in that. If it is run such that

at startup the * standard input or output are associated with a file or pipe, or under X the * variable `DISPLAY` is not set it will try to start up in console * mode. The flag `-w` indicates that the system should run in console * more regardless, while `-w+` attempts a window even if that seems * doomed to failure. When running the system to obey a script it will often make * sense to use the `-w` option. Note that on Windows the system is * provided as two separate (but almost identical) binaries. For example the * file `cs1.exe` is linked in windows mode. A result is that if * launched from the command line it detaches from its console, and if launched * by double-clicking it does not create a console. It is in fact very ugly when * double clicking on an application causes an unwanted console window to appear. * In contrast `cs1.com` is a console mode version of just the same * program, so when launched from a command line it can communicate with the * console in the ordinary expected manner. * * The option is in fact processed at an earlier stage than here if windowing * is possible at all!

- x * `-x` is an option intended for use only by system * support experts – it disables trapping if segment violations by * `errorset` and so makes it easier to track down low level disasters – * maybe! This can be valuable when running under a debugger since if the * code traps signals in its usual way and tries to recover it can make it a lot * harder to find out just what was going wrong.
- y * `-y` is at present unused.
- z * When bootstrapping it is necessary to start up the system for one initial time * without the benefit of any image file at all. The option `-z` makes * this happen, so when it is specified the system starts up with a minimal * environment and only those capabilities that are present in the CSL * kernel. It will normally make sense to start loading some basic Lisp * definitions rather rapidly. The files `compat.lsp`, * `extras.lsp` and `compiler.lsp` have Lisp source for the * main things I use, and once they are loaded the Lisp compiler can be used * to compile itself.

3 Predefined variables

!!fleps1

There is a function `safe!-fp!-plus` that performs floating point arithmetic but guarantees never to raise an exception. This value was at one stage related to when small values created there got truncated to zero, but the current code does not use the Lisp variable at all and instead does things based on the bitwise representation of the numbers.

!\$cslbase

* See **!@cslbase**.

!\$eof!\$

The value of this variable is a pseudo-character returned from various read functions to signal end-of-file.

!\$eol!\$

The value of this variable is an end-of-line character.

!\$ff!\$

The value of this variable is a form-feed character.

!*applyhook!*

If this is set it might be supposed to be the name of a function used by the interpreter as a callback, but at present it does not actually do anything!

!*break!-loop!*

If the value of this is a symbol that is defined as a function of one argument then it is called during the processing on an error. This has not been used in anger and so its whole status may be dubious!

!*carcheckflag

In general CSL arranges that every **car** or **cdr** access is checked for validity. Once upon a time setting this variable to nil turned such checks off in the hope of gaining a little speed. But it no longer does that. It may have a minor effect on array access primitives.

!*comp

When set each function is compiled (into bytecodes) as it gets defined.

!*debug!-io!*

An I/O channel intended to be used for diagnostic interactions. The concept and name is taken from Common Lisp, but there is in fact no real separation between this and the standard input and output streams.

!*echo

When this is non-nil characters that are read from an input file are echoed to the standard output. This gives a more complete transcript in a log file, but can sometimes amount to over-verbose output.

!*error!-messages!*

Has the value nil and does not do anything! At one stage the idea had been that CSL's error messages would be in this table so as to be available from Lisp, but that never happened.

!*error!-output!*

An I/O channel intended for diagnostic output. The concept comes from Common Lisp but Standard Lisp (and hence CSL) does not really exploit it.

!*evalhook!*

See **!*applyhook!***. This also does not do anything at present, but it did it would be a place to put the name of a function that would be called by the interpreter when applying a function.

!*gc!-hook!*

If this is set to have as its value that is a function of one argument then that function is called with **nil** on every minor entry to the garbage collection, and with argument **t** at the end of a “genuine” full garbage collection. This may sometimes be of interest for those who want to notice when garbage collection happens but want to control how they are informed rather than relying on the displayed text that the **verbos** function controls.

!*lower

In CSL if the fluid variable **!*lower** is set (which it is by default) then when characters are read they are folded to lower case. The related variable **!*raise** causes input to be folded to upper case on input. In the original Standard Lisp the true internal names of all functions were in upper case, and **!*raise** was used so that code using it could be written in mixed or lower case. At some stage it was accepted that upper case was a throw-back to the days of FORTRAN and punched cards, so the CSL switched to using lower case internally and the **!*lower** flag allowed the existing Reduce sources to survive. At some time in the future I hope that Reduce will arrange to have both ***raise** and **!*lower** set to **nil** so that it becomes a case-sensitive system.

Note that the PSL Lisp system once (but not now) used **!*lower** to cause internally upper case symbols to be printed in lower case, so that there it is a control of output rather than input case folding. PSL has functions **input!-case** and **output!-case** that are relevant in this respect. So code that is to be portable between the two Lisps needs to take care.

!*macroexpand!-hook!*

Common Lisp would like it to be possible to specify a function that would be called to allow overriding of the normal method of expanding macros. This variable exists in CSL in case at any stage a serious need for this capability arises, but at present any use of this variable has no effect.

!*math!-output!*

In the case that CSL is being used with Reduce and its “fancy” maths mode display is available in a GUI then this provides a special output stream that displayable layout information in a T_EX-like notation that is not documented here because it is potentially in the process of being updated. The `tmprint` package in Reduce generates this material and collaborates with the mathematical display parts of CSL.

!*package!*

* Internally CSL stores the table that maps names into symbols in a way * following the style of the Common Lisp package system. When used as * a Standard Lisp there is only one package and no distinction between * internal and external name visibility, but this name provides access to * the main tdata-structure involved. It is in general expected that this * will be used via the `oblist` function, not directly by users.

!*pgwd

* See **!*plap**.

!*plap

* When the CSL compiler runs to generate byte-codes if **!*plap** or **!*pgwd** is set then the generated code is displayed. This may * be of interest when debugging or for anybody who wants to explore the * Lisp bytecode model that is used.

!*pretty!-symmetric

* The `prettyprint` function displays a Lisp expression neatly * indented. If this variable is set (which by default it is) symbols and * strings are shown with escape characters and quotation marks so that the * indented form could be re-input. If this variable is set to `nil` * that does not happen – the output may not be re-readable by CSL but in * some cases it may be easier for a human reader to decipher.

!*prinl!-fn!*

* Used internally by the functions `prinl` and `princl` * that can print data structures that are re-entrant or looped. Not for use * by end-users.

!*prinl!-index!*

* Used internally by the functions `prinl` and `princl` * that can print data structures that are re-entrant or looped. Not for use * by end-users.

!*prinl!-visited!-nodes!*

* Used internally by the functions `prinl` and `princl` * that can print data structures that are re-entrant or looped. Not for use * by end-users.

!*print!-array!*

* In `prin1` if this variable is `nil` arrays and * structures are not printed in full. In some cases this merely loses valuable * information, while in others it leads to output that is more concise and * legible and hence more useful.

!*print!-length!*

* In `prin1` if this variable is set to an integer then that specifies * the largest number of items in a list that will be displayed.

!*print!-level!*

* In `prin1` if this variable is set to an integer then that specifies * the greatest depth of nesting of lists before the printing gives up. This * and **!*print!-length!*** may very occasionally be useful when * faced with huge lists of which only the top few layers are relevant.

!*pwrds

* This is normally set, and it causes the compiler to display a message * commenting on how many bytes were used in the compiled version of each * function that is processed.

!*query!-io!*

* An I/O channel intended to be used for query interactions. The concept * and name is taken from Common Lisp, but there is in fact no real separation * between this and the standard input and output streams.

!*quotes

* Used in the prettyprinter to determine whether the form `(quote x)` * should be displayed as `'x`. By default it is.

!*raise

* See **!*lower**.

!*redefmsg

* If this is set a message is displayed when a function is redefined.

!*resources!*

* See the `resource!-limit` function.

!*savedef

* If this variable is set then when you define a function and compile it the * original interpretable Lisp form of the definition is saved under the * property-name **!*savedef** so that it could be recovered using * `get`. If the function is being compiled into a fasl-file for * later reloading the lisp form of the definition is saved there so that when * `load!-module` or `load!-source` is used it can be * retrieved. This

facility is activated when the “bootstrap” version of * Reduce is built so that in effect the full source code is available at * run-time. The availability of source in that way can be useful for forms * of global analysis or optimisation of the code – for instance Reduce * uses it to find the definitions of functions that it wants to optimise * int C code rather than the slower (but more compact) bytecodes it uses * for most things.

!*spool!-output!*

* The `spool` function or the command-line option `-l` * can establish a file that normal output is copied to as a log. This variable * holds a handle to that file.

!*standard!-input!*

* Standard Lisp specifies that to select input from the “standard” source * one goes (`rds nil`). In CSL this is underpinned by having an * input stream as stored in this variable following the naming convention used * by Common Lisp.

!*standard!-output!*

* As **!*standard!-input!*** but for output.

!*terminal!-io!*

* A Common Lisp motivated variable which is intended to provide access to * the “terminal”. In Standard Lisp you are expected to use `rds` * and the precise concept of a terminal is not really defined.

!*trace!-output!*

* The Lisp trace facility tends to send output to this which is a synonym * for the original standard output.

!@cslbase

* This variable is not actually predefined, but I will nevertheless give some * explanation of how it is used during the bootstrapping process that makes * a CSL or a Reduce image. When you attempt to open a file you may give * a path starting with an initial “\$word/...” or “\${word}/...”. * these notations of course model typical Unix-style parameter substitution. * The expansion proceeds by first checking if a Lisp variable “@word” * exists with a string or a symbol as its name. If so that value is used * as the expansion. If that scheme fails the system next looks for an * environment variable and uses its value. This the use of a Lisp variable * “@word” takes priority over the system environment. Finally if there is * no environment variable available a Lisp variable with name “\$word” is * checked and if its value is a string or symbol that is the expansion, * otherwise the expansion will be empty. * * This is used in the build sequences by

passing a command-line * option `-D@cslbase=...` that predefines * `@cslbase` to refer to a key directory where necessary files * can be found. * By defining this rather than `$cslbase` there is no chance that * any odd values in the (shell) environment will cause trouble, and the * lines such as `(rdf "$cslbase/compat.lsp")` can appear in the build * code without any need for absolute path names or any * reliance on the setting of a current directory. * * In a file-name a path that starting `~/` or * `~name/` tries to identify the home directory of the * current or named user. This has a clear-cut meaning on Unix-like * platforms but should perhaps not be relied on under Windows.

blank

The value of this variable is an space or blank character. This might otherwise be written as `"! "`.

carriage!-return

The value of this variable is a carriage-return character that could arise in DOS-format files.

common!-lisp!-mode

* The CSL Lisp system was designed so that if necessary much of the code could * be shared with a version that met the Common Lisp Standard. At no stage has * there been anything like a complete Common version. This is both because the * main use of CSL has been to support Reduce and that wants Standard (not * Common) Lisp, and because providing complete support for all the * functionality in Common Lisp would be a lot of work and would tend to make * the code bulkier (and hence necessarily less reliable) and slower. However * various key underpinnings for Common Lisp are present in the C-coded sources, * generally guarded by `"#ifdef COMMON"`. If the Lisp has been * built in this way then this variable will be set in order that users can * readily detect the situation. At one stage the Axiom algebra system could * be built using the limited Common Lisp compatibility mode, but the recent * Open Source versions of Axiom have probably changed leaving that not an * easy option.

crbuf!*

* This is a variable used by the Reduce parser, and as a matter of caution it * is to be treated as reserved in the Lisp system.

emsg!*

* After a call `(error nn msg)` this variable gets set to the * value of the second argument (`msg`). This may help if you have had * a failure and want to see if it resulted from a call to the `error` * function and if so what message had been used with it!

eof!*

* Used in Reduce in association with detecting and handling end-of-file * conditions, and reserved in the Lisp to avoid potential interference with * that.

esc!*

The value of this variable is the character “escape”. As a non-printing character use of this is to be viewed as delicate.

lispsystem!*

* This variable is initialised at the start of any run of Lisp to hold * information about the computer in use and the collection of features * available in the Lisp. The items that might be present are explained * further in Section 4.

load!-source

* The function **load!-source** will load data from a fasl file * and is intended to make it possible to have saved uncompiled Lisp forms * for functions there – and to be able to reload then in a selective * way. The **load!-source** variable can tune this behaviour. See the * explanation of the function for further details.

nil

* **nil** is the fundamental Lisp atom used to stand for “false”, * used to terminate lists and generally something that every Lisp programmer * will already understand about. In Standard Lisp **nil** is a * symbol and as such you may not take **car** or **cdr** of * it. So any code that had been developed for Common Lisp and relies on being * able to treat it as if it was a non-empty list will need revision. CSL * always checks for valid access so not only would (**car nil**) be * a mistake, but any attempt to do it will lead to an exception being * raised. **nil** must not be used as a name of an argument or a * **prog** variable or in any other context that could attempt to * alter its value.

ofl!*

* Used in Reduce in association with tracking output files, and reserved in * CSL to avoid conflict with that.

program!*

* Used by the Reduce parser to hold a command that has just been parsed, and * reserved in CSL in order to ensure that there is no conflict with that.

s!:bn

* Used internally by the prettyprint function.

`s!:bufferi`
 * Used internally by the prettyprint function.

`s!:bufferp`
 * Used internally by the prettyprint function.

`s!:gensym!-serial`
 * internal variable used by `dated!-name`.

`s!:indblanks`
 * Used internally by the prettyprint function.

`s!:indentlevel`
 * Used internally by the prettyprint function.

`s!:initialblanks`
 * Used internally by the prettyprint function.

`s!:lmar`
 * Used internally by the prettyprint function.

`s!:pendingrpars`
 * Used internally by the prettyprint function.

`s!:rmar`
 * Used internally by the prettyprint function.

`s!:rparcount`
 * Used internally by the prettyprint function.

`s!:stack`
 * Used internally by the prettyprint function.

`t`
 * The Lisp value that stands for “true”. Any attempt to reset or *
 rebind `t` will be an error. The value of `t` is * itself. Yoy may recall
 that in Lisp 1.5 the value of `t` * had been `*t*`, but Standard Lisp
 mandates the behaviour * implemented here. As far as truth values
 are concerned, `nil` * is treated as meaning “false” and anything that
 is non-`nil` * is true, including as an obvious special case `t`.

`tab`
 The value of this variable is a tab character.

`thin!*`
 * In the prettyprinter if `thin!*` values (default 5) can be * fitted on a
 single line. The idea behind this is so that long lists can * in relevant
 cases be displayed almost horizontalloy rather than * vertically, as in
 *

```

* (one two three four five          (one
*   six seven eight nine ten   vs.   two
*   eleven twelve)              three
*                                  ...
*

```

ttype!*

* Used by the Reduce parser, and so best not used for other purposes.

4 Items that can appear in **lispsystem!***

* * There is a global variable called **lispsystem!*** whose value is * reset in the process of CSL starting up. An effect of this is that if the * user changes its value those changes do not survive a preserving and * re-loading a heap image: this is deliberate since the heap image may be * re-loaded on a different instance of CSL possibly on a quite different * computer or with a different configuration. The value of * **lispsystem!*** is a list of items, where each item is either an atomic tag * of a pair whose first component is a key. In general it would be unwise * to rely on exactly what information is present without review of the code * that sets it up. The information may be of interest to anybody but some tags * and keys are reflections of experiments rather than fully stable facilities. *

(c!-code . count) * This will be present if code has been optimised into C through the source * files u01.c to u60.c, and in that case the value tells you how many functions * have been optimised in this manner. *

common!-lisp * For a project some while ago a limited Common Lisp compatibility mode was * being developed, and this tag indicated that it was active. In that case all * entries are in upper case and the variable is called ***FEATURES*** * rather than **lispsystem!***. But note that this Lisp has never even * aspired to be a full Common Lisp, since its author considers Common Lisp to * have been a sad mistake that must bear significant responsibility for the * fact that interest in Lisp has faded dramatically since its introduction. *

cs1 * A simple tag intended to indicate that this Lisp system is CSL and not any * other. This can of course only work properly if all other Lisp systems * agree not to set this tag! In the context of Reduce I note that the PSL * Lisp system sets a tag **ps1** on **lispsystem!*** and * the realistic use of this is to discriminate between CSL and PSL hosted * copies of Reduce.

debug * If CSL was compiled with debugging options this is present, and one can imagine * various bits of code being more cautious or more verbose if it is detected.

(executable . name) * The value is the fully rooted name of the executable file that was launched.

fox * Used to be present if the FOX GUI toolkit was detected and incorporated as * part of CSL, but now probably never used!

(name . name) * Some indication of the platform. For instance on one system I use it * is linux-gnu:x86_64 and on another it is just win32.

(opsys . operating-system) * Some crude indication of the host operating system.

operating system identity * The name of the current operating system is put on the list. Exactly what * form is not explicitly defined!

pipes * In the earlier days of CSL there were computers where pipes were not * supported, so this tag notes when they are present and hence the facility * to create sub-tasks through them can be used.

record_get * An extension to the CSL profiling scheme it is possible to compile * a special version that tracks and counts each use of property-list access * functions. This can be useful because there are ways to give special * treatment to a small number of flags and a small number of properties. The * special-case flags end up stored as a bitmap in the symbol-header so avoid * need for property-list searching. But of course recording this extra * information slows things down. This tag notes when the slow version is * in use. It might be used to trigger a display of statistics at the end of * a calculation.

reduce * This is intended to report if the initial heap image is for Reduce rather than * merely for Lisp.

(revision . ver) * The CSL subversion revision number.

(shortname . name) * Gives the short name of the current executable, without its full path.

showmath * If the “showmath” capability has been compiled into CSL this will be present * so that Lisp code can know it is reasonable to try to use it.

showmath1 * This marks the fact that this version of CSL will support the output * style where flat simple text precedes TeX output, with a U+0003 (end of text) * marker to separate.

sixty!-four * Present if the Lisp was compiled for a 64-bit computer.

termed * Present if a cursor-addressable console was detected.

texmacs * Present if the system was launched with the `--texmacs` flag. *
The intent is that this should only be done when it has been launched
with * texmacs as a front-end.

win32, win64 * Any windows system puts **win32** in `lispsystem!*`. * If
64-bit windows is is use then **win64** is also included

windowed * Present if CSL is running in its own window rather than in
console mode.

5 Flags and Properties

Most of tags here are probably not much use to end-users, but I am noting
them as a matter of completeness.

lose * If a name is flagged as `ttfamily lose` then a subsequent attempt to *
define or redefine it will be ignored.

s!:ppchar and **s!:ppformat** * These are used in the prettyprint code found
in `extras.red`. A * name is given a property **s!:ppformat** if in pret-
typrinted display * its first few arguments should appear on the same
line as it if at all * possible. The **s!:ppchar** property is used to make
the display of * bracket characters a little more tidy in the source code.

switch * In the Reduce parser some names are “switches”, and then di-
rectives such * as `on xxx` and `off xx` have the effect of setting or *
clearing the value of a variable `!*xxx`. This is managed by * setting
the **switch** flag om `xxx`. CSL sets some * things as switches ready for
when they may be used by the Reduce parser.

!~magic!-internal!-symbol!~ * CSL does not have a clear representa-
tion for functions that is separated from * the representation of an
identifier, and so when you ask to get the value * of a raw function
you get an identifier (probably a gensym) and this * tag is used to link
such values with the symbols they were originally * extracted from.

6 Functions and Special Forms

* * Each line here shows a name and then one of the words *expr*, * *fexpr* or
macro. In some cases there can also be special * treatment of functions by
the compiler so that they get compiled in-line. *

abs *expr*

* This takes one argument that should be a number – and returns its
absolute * value. In Common Lisp mode it would find the magnitude

of a complex * number, but in normal Standard Lisp mode the only cases that arise are * integers are floating point values.

`binary_close_input expr`
* Not yet written

`binary_close_output expr`
* Not yet written

`binary_open_input expr`
* Not yet written

`binary_open_output expr`
* Not yet written

`binary_prin1 expr`
* Not yet written

`binary_prin2 expr`
* Not yet written

`binary_prin3 expr`
* Not yet written

`binary_prinbyte expr`
* Not yet written

`binary_princ expr`
* Not yet written

`binary_prinfloat expr`
* Not yet written

`binary_read2 expr`
* Not yet written

`binary_read3 expr`
* Not yet written

`binary_read4 expr`
* Not yet written

`binary_readbyte expr`
* Not yet written

`binary_readfloat expr`
* Not yet written

`binary_select_input expr`
* Not yet written

`binary_terpri expr`
* Not yet written

`bps!-getv expr`
* Not yet written

`bps!-putv expr`
* Not yet written

`bps!-upbv expr`
* Not yet written

`break!-loop expr`
* Not yet written

`c_out expr`
* Not yet written

`caaaaar expr`
see `caar`.

`caaadr expr`
see `caar`.

`caaar expr`
see `caar`.

`caadar expr`
see `caar`.

`caaddr expr`
see `caar`.

`caadr expr`
see `caar`.

`caar ...cddddr expr`
Names that start with `c`, then have a sequence of `a` or `d`s and finally `r` provide shorthand functions for chains of uses of `car` and `cdr`. Thus for instance `(cadar x)` has the same meaning as `(car (cdr (car x)))`.

`cadaar expr`
see `caar`.

`cadadr expr`
see `caar`.

`cadar expr`
see `caar`.

`caddar expr`
see `caar`.

`cadddr expr`
see `caar` and `fourth`.

`caddr expr`
see `caar` and `third`.

`cadr expr`
see `caar` and `second`.

`car expr`
For a non-empty list the function `car` will return the first element. For a dotted pair (created using `cons`) it extracts the first component. This is the fundamental low-level data structure access function in Lisp. See `cdr` for the function that returns the tail or a list or the second component of a dotted pair. In CSL any attempt to take `car` of an atom should be detected and will be treated as an error. If CSL had been compiled in Common Lisp mode (which is now not probable) a special exemption would apply and `car` and `cdr` of the empty list `nil` would be `nil`.

`car!* expr`
This function behaves like `car` except that if its argument is atomic then the argument is returned unaltered rather than that case being treated as an error.

`cdaaar expr`
see `caar`.

`cdaadr expr`
see `caar`.

`cdaar expr`
see `caar`.

`cdadar expr`
see `caar`.

`cdaddr expr`
see `caar`.

`cdadr expr`
see `caar`.

`cdar expr`
see `caar`.

`cddaar expr`
see `caar`.

`cddadr expr`
see `caar`.

`cddar expr`
see `caar`.

`cdddar expr`
see `caar`.

`cddddr expr`
see `caar`.

`cdddr expr`
see `caar`.

`cddr expr`
see `caar`.

`cdr expr`
See `car`.

`char!-code expr`
* Not yet written

`char!-downcase expr`
* Not yet written

`char!-upcase expr`
* Not yet written

`check!-c!-code expr`
* Not yet written

`cl!-equal expr`
* Not yet written

`close!-library expr`
* Not yet written

`code!-char expr`
* Not yet written

`compile!-all expr`
* Not yet written

`convert!-to!-evector expr`
* Not yet written

`copy!-module expr`
* Not yet written

`create!-directory expr`
* Not yet written

`dated!-name expr`
* Not yet written

`define!-in!-module expr`
* Not yet written

`delete!-file expr`
* Not yet written

`delete!-module expr`
* Not yet written

`do!* macro`
* Not yet written

`double!-execute expr`
* Not yet written

`enable!-backtrace expr`
* Not yet written

`enable!-errorset expr`
* Not yet written

`eq!-safe expr`
* Not yet written

`eval!-when fexpr`
* Not yet written

`file!-length expr`
* Not yet written

`file!-readablep expr`
* Not yet written

`file!-writeablep expr`
* Not yet written

`flagp!*! expr`
* Not yet written

`fp!-evaluate expr`
* Not yet written

```

funcall!* expr
    * Not yet written

get!* expr
    * Not yet written

get!-current!-directory expr
    * Not yet written

get!-lisp!-directory expr
    * Not yet written

hash!-table!-p expr
    * Not yet written

hashtagged!-name expr
    * Not yet written

input!-libraries fexpr
    * Not yet written

instate!-c!-code expr
    * Not yet written

internal!-open expr
    * Not yet written

is!-console expr
    * Not yet written

let!* fexpr
    * Not yet written

library!-members expr
    * Returns a list of all the modules that could potentially be loaded
    using * load!-module. See list!-modules to get a human * read-
    able display that looks more like the result of listing a directory, or *
    modulep for checking the state of a particular named module. *

library!-name expr
    * Not yet written

list!* fexpr
    * Not yet written

list!-directory expr
    * Not yet written *

```

```

list!-modules expr
  * This prints a human-readable display of the modules present in the
  * current * image files. This will include "InitialImage" which is the
  * heap-image * loaded at system startup. For example *

      * > (list!-modules)
      *
      * File d:\csl\csl.img (dirsize 8 length 155016, Writable):
      *   compat      Sat Jul 26 10:20:08 2008 position 556 size: 9320
      *   compiler     Sat Jul 26 10:20:08 2008 position 9880 size: 81088
      *   InitialImage Sat Jul 26 10:20:09 2008 position 90972 size: 64040
      *
      * nil
      *

  * * See library!-members and modulep for functions that * make it
  * possible for Lisp code to discover about the loadable modules that are
  * available.

list!-to!-string expr
  * Not yet written

list!-to!-symbol expr
  * Not yet written

list!-to!-vector expr
  * Not yet written

list2!* expr
  * Not yet written

list3!* expr
  * Not yet written

load!-module expr
  * Not yet written

load!-source expr
  * Not yet written

lose!-precision expr
  * Not yet written

macro!-function expr
  * Not yet written

macroexpand!-1 expr
  * Not yet written

make!-bps expr
  * Not yet written

```

`make!-function!-stream expr`
* Not yet written

`make!-global expr`
* Not yet written

`make!-random!-state expr`
* Not yet written

`make!-simple!-string expr`
* Not yet written

`make!-special expr`
* Not yet written

`math!-display expr`
* Not yet written

`member!*** expr`
* Not yet written

`rplaca expr`
This is a destructive function in that it alters the data structure that it is given as its first argument by updating its `car` component. The result is the updated object. See `rplacd` for the corresponding function for updating the `cdr` component.

`rplacd expr`
See `rplaca`

`!~block fexpr`
* Not yet written

`!~let fexpr`
* Not yet written

`!~tyi expr`
* Not yet written